D07 · Cross-Docking & Flow-Through

02 · Yard management & dock scheduling

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1 WHAT IT IS

Yard management tracks every trailer on site — identity, contents, status and dwell — while dock appointment scheduling controls when carriers arrive and which door they use.

2 WHY IT MATTERS

Most dock congestion is arrival clustering, not a shortage of doors. **When carriers arrive whenever they choose, trucks queue in the morning while docks sit idle in the afternoon.** Long turnaround also has an external cost: carriers price the risk of waiting into their rates, so a slow facility pays for it in every freight negotiation.

3 HOW IT IS DONE

1. **Make appointments mandatory.** Booked windows for every carrier; unbooked arrivals served last.
2. **Track trailers in a YMS.** Identity, contents, status and dwell time visible to the dock team.
3. **Allocate doors by flow type.** Cross-dock and urgent loads get priority over storage receipts.
4. **Score carriers on window compliance.** Add it to the carrier scorecard alongside OTIF and damage.

IN PRACTICE

Appointment scheduling is worth more than additional dock doors

Most dock congestion is arrival clustering, not insufficient doors. Spreading arrivals across the working day is a scheduling change; adding doors is capital. Exhaust the first before proposing the second.

Put carrier window compliance on the carrier scorecard

A booked window with no consequence for missing it is a suggestion. Window compliance belongs alongside OTIF and damage rate in the D06 carrier scorecard — the same governance, the same volume consequences.

Allocate docks by flow type — not by arrival order

A cross-dock load queuing behind a storage receipt breaks a synchronized flow to serve one that could have waited. Dock priority should follow the cost of delay, and first-come-first-served ignores it entirely.

Measure turnaround by hour of day — the average hides the peak

An acceptable daily mean can contain a three-hour window where turnaround triples. Carriers experience the peak, not your average — and they price accordingly.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Yard and dock scheduling — DC with chronic morning congestion — Carriers arrive unannounced, concentrated in the early morning. Docks sit idle in the afternoon while trucks queue at 7am.

Root cause	Mechanism	Countermeasure
No appointment discipline — carriers arrive at their convenience	Arrivals cluster in a 3-hour morning window	Mandatory booked windows · Unbooked arrivals served last, not first
No trailer visibility — nobody knows what is on site	Loaded trailers wait behind empties · Manual yard checks	YMS with trailer identity, contents and dwell time

Root cause	Mechanism	Countermeasure
Dock assignment is first-come-first-served	Cross-dock loads queue behind low-priority storage receipts	Dock allocation by flow type · Cross-dock loads get window priority
Carrier window compliance is never measured	No consequence for missing a booked slot	Window compliance on the carrier scorecard

Result (illustrative): The congestion was a scheduling problem, not a dock count problem. Spreading arrivals across the working day and prioritizing cross-dock loads relieved the morning queue without adding a single dock door.

HOW IT IS MEASURED

Truck Turnaround Time (gate-in to gate-out)

Truck Turnaround TimeAverage elapsed time from truck gate-in to gate-out

Benchmark: Track by flow type and by hour of day · The distribution matters more than the average — a good mean can hide a 3-hour morning peak Long turnaround has a cost you do not see on your own P&L: carriers price detention risk into their rates. A DC known for slow turnaround pays for it in every freight negotiation.

Appointment Compliance % (arrivals inside the booked window)

Appointment Compliance %(Arrivals within the booked dock window / Total booked arrivals) × 100

Benchmark: High enough that the schedule is trustworthy · Measure by carrier, not only in aggregate Appointment compliance is the input that makes every downstream plan feasible. A cross-dock schedule, a labor plan, and a dispatch wave all assume the truck arrives when it said it would.

WHAT YOU WOULD USE

Yard & Dock

C3 Solutions / Opendock / FourKites Dynamic Yard — Module: YMS + Carrier Appointment SchedulingDedicated yard management and dock appointment platforms with carrier-facing booking portals, trailer tracking and dwell time reporting.

SAP EWM — Yard Management — Module: Yard Management + Door AssignmentSAP EWM yard module with trailer check-in/check-out, yard location management and door assignment integrated with the warehouse process.

4 SO WHAT

Schedule arrivals before adding doors. Mandatory appointments, trailer visibility, door priority by flow and carrier scorecards that count window compliance.

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